

```
$ /usr/local/nagios/libexec/check_nrpe -H <REMOTE-HOST-IP-ADDRESS>
NRPE v2.15
```

The output should display the NRPE version installed on the remote host, as shown above.

Adding hosts to Nagios monitoring: In order to add new servers to Nagios monitoring, let's create a directory with name servers inside the `/usr/local/nagios/etc/objects` directory. Configuration files for all the servers we add will reside in this directory.

```
$ mkdir /usr/local/nagios/etc/objects/servers
```

In order to add this directory in the configuration file, edit the `/usr/local/nagios/etc/nagios.cfg` file and uncomment the following line:

```
# cfg_dir=/usr/local/nagios/etc/servers
```

Now, change to the 'servers' directory and create a new configuration file for our new Nagios client. Add the following details to the configuration file, for which you can take help from the `templates.cfg` file:

```
$ cd /usr/local/nagios/etc/servers
$ vi MyLinuxServer.cfg
define host {
    name                MyLinuxServer
    address              192.168.0.110
    use                  generic-host
    check_period         24x7
    check_interval       5
    retry_interval       1
    max_check_attempts   5
    check_command        check-host-alive
    notification_period   workhours
    notification_interval 60
    contact              nagiosadmin
    register             1
}
```

Adding services for the hosts: To add any service to the hosts, one might need to create a command to monitor that service. This can be done by editing the `/usr/local/nagios/etc/objects/commands.cfg` file.

```
define command{
    command_name check_nrpe
    command_line $USER1$/check_nrpe -H $HOSTADDRESS$ -c $ARG1$
}
```

Similar to the hosts, we should create a directory called

'services' in order to keep custom services configuration files. Again, the necessary changes should be made to the `/usr/local/nagios/etc/nagios.cfg` file.

```
$ mkdir /usr/local/nagios/etc/objects/services
$ echo "cfg_dir=/usr/local/nagios/etc/services" >> /usr/local/nagios/etc/nagios.cfg
```

To add services for the host `MyLinuxServer`, let's create a configuration file `MyLinuxServer.cfg` in the `services` directory as shown below:

```
$ cd /usr/local/nagios/etc/objects/services
$ vi MyLinuxServer.cfg
define service {
    use                generic-service
    host_name          MyLinuxServer
    service_description Server Uptime
    check_command       check_nrpe!check_uptime

    notifications_enabled 1
}
```

Note: Whenever changes are made to the `/usr/local/nagios/etc/nagios.cfg` file, execute the following command to check for any errors in the configuration:

```
/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
```

After adding the host to the Nagios configuration, restart the Nagios service once on the Nagios server, and observe the changes in the Nagios Web console (Figure 6).

In a similar manner, many more hosts and services can be added to the Nagios configuration.

Creating custom Nagios plugins using the Bash script:

Even though the Nagios plugins that we have already installed offer a wide variety, one might need to create one's own Nagios plugin in order to monitor a specific service. The Nagios plugin supports a wide range of programming languages including Bash, Python, Perl, C/C++, etc. But one common fact that needs to be remembered while creating a Nagios plugin using any of the mentioned programming languages is the syntax of the output, which is as follows:

```
<STATUS> - <DATA TO BE DISPLAYED ON WEB CONSOLE> | <DATA NEEDED FOR GENERATING GRAPHS>
```



Figure 6: Nagios host added